SORBONNE UNIVERSITY
Master 2 IMA
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Report PRAT Project

Video Interpolation for Geospatial data

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Abstract

Temporal video interpolation of geospatial data presents some challenges that most of computer vision techniques dependent on the assumptions on rigid-body motion fail to address. In this report we detail the implementation of **STint (Self-supervised Temporal Interpolation)** (1), a novel framework designed to enhance the temporal resolution of climate datasets without the need for ground-truth labels or even motion information.

The methodology is based on a dual cycle consistency constraint, which forces the model to learn the underlyiing temporal evolution of objects by reconstructing original frames through 2 stages predictions. The model architecture uses a U-Net enhanced with Squeeze-and-Excitation (SE) blocks to better model interdependencies among convolutional features.

Chapter 1

State of the art

1.1 Simple Interpolation Methods

1.1.1 Persistence (Zero-Order)

he simplest method, which assumes the signal remains constant until the next sample. The intermediate frame is an exact duplicate of the preceding frame.

1.1.2 Linear Interpolation (First-Order)

This method assumes a constant, straight-line change between frames. It blends the two nearest frames in proportion to the time gap (e.g., $F_{interp} = \frac{1}{2}F_n + \frac{1}{2}F_{n+1}$).

1.1.3 Cubic Interpolation (Second-Order)

A higher-order approach that uses a third-degree polynomial to fit a smooth curve through a sequence of known frames.

1.2 Optical Flow-Based Methods

Optical flow tracks the apparent motion of pixels between two frames, represented as a 2D vector field.

- **Mechanism**: Models such as Super SloMo utilize U-Net architectures to estimate bi-directional flow, which is then used to warp input frames into an intermediate position.

- **Limitations**: While effective for high-frame-rate videos with rigid objects, these methods struggle with the non-rigid deformations.

1.3 Diffusion-Based Models

Diffusion models leverage probabilistic generative processes to reconstruct frames through iterative denoising.

- **Capability**: Methods like VIDIM use cascaded diffusion models to generate sequences of intermediate frames, which are then refined for high resolution.

- **Pros and Cons**: These models are excellent at handling complex, non-linear, or ambiguous motions. However, they suffer from slow inference speeds due to the iterative nature of the denoising process.

Introduction

The goal of video interpolation is the generate frames between existing frames to increase the temporal resolution. By enhancing temporal resolution, we can move from a sparse series of data points to continuous stream. This can be used for patching holes in historical records or recovering data during sensor downtime...

1.4 Context and motivation: The importance of temporal upsampling in climate modeling and satellite imagery

In the context of geospatial analysis, temporal upsampling is crucial for transforming low-resolution climate models into high-resolution data. For example, data recorded at 3-hourly intervals can be interpolated to 1-hourly data to provide a more granular view of atmospheric or land variables.

1.5 Problem Statement

Traditional video interpolation methods predominantly rely on optical flow, which tracks the motion of pixels between frames. While effective for high-frame-rate videos where objects move rigidly, these methods fail when applied to geospatial data for several reasons:
- Low temporal granularity: Geospatial measurements are often taken in intervals of hours or days, unlike video frames which are captured in fractions of a second.
- Non-rigid derformations: Elements like clouds, atmospheric fronts, and ocean currents undergo complex deformations that defy the rigid-body assumptions of optical flow.

1.6 Project Objectives

The first objective of this project is to implement and evaluate **STint**. Our goal is to demonstrate that this approach generates intermediate frames that are more physically consistent and visually closer to ground truth than traditional baselines on geospatial variables like earth temperature.

Chapter 2

Methodology

The core of this method lies in the dual cycle consistency framework which enables the network to learn temporal interpolation in a fully unsupervised manner. This approach removes the need for ground-truth labels by enforcing a reconstruction constraint across multiple stages of the sequence.

2.1 Dual Cycle Consistency Framework

The fundamental idea of cycle consistency is that if a system undergoes a transformation followed by its inverse, it should return to its original starting point. In the context of STint, we expand this into a two-stage process to regularize the model and ensure temporal stability.

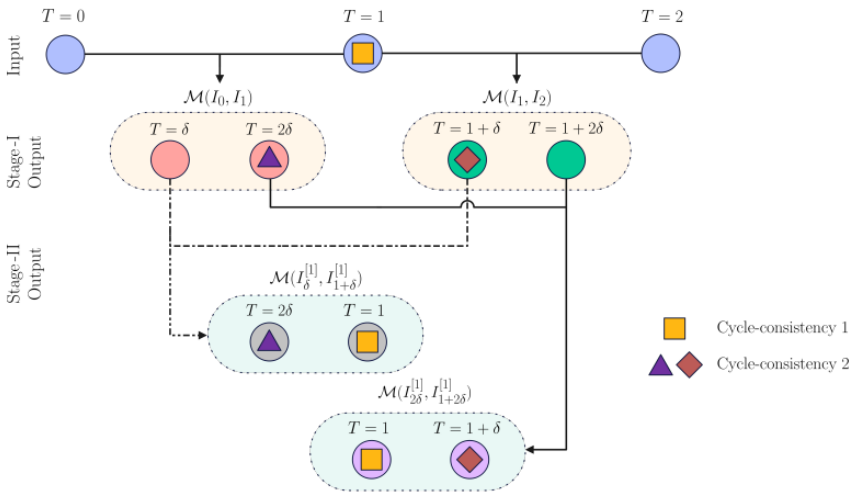


Figure 2.1: Dual cycle consistency. Source : (1)

2.1.1 Stage I: Initial Interpolation

Given a triplet of consecutive frames $[I_0, I_1, I_2]$, the goal is to train a neural network $\mathcal{M}$ to predict intermediate frames. In the first stage, the network takes two pairs of consecutive frames ($[I_0, I_1]$ and $[I_1, I_2]$) and generates two intermediate frames for each pair:
- Pair 1: $[I_\delta^{[1]}, I_{2\delta}^{[1]}] = \mathcal{M}(I_0, I_1)$
- Pair 2: $[I_{1+\delta}^{[1]}, I_{1+2\delta}^{[1]}] = \mathcal{M}(I_1, I_2)$

2.1.2 Stage II: Reconstruction and Consistency

In the second stage, we use the predicted intermediate frames from Stage 1 as inputs to the same network $\mathcal{M}$ to reconstruct the original sequence. This stage is critical because it creates the "cycle necessary" for unsupervised learning.

The framework employs two specific losses to optimize the network:
- Loss 1: This loss ensures that the model can reconstruct the original middle frame (I_1). The network takes the predicted frames at $T = \delta$ and $T = 1 + \delta$ (and also those at 2δ ans $1 + 2\delta$). The error between this reconstruciton and the original input I_1 serves as the primary training signal.

$$LLC_1 = argmin_{M_\theta}(||I_1 - \mathcal{M}(I_\delta^{[1]}, I_{1+\delta}^{[1]})(2)||_1 + ||I_1 - \mathcal{M}(I_{2\delta}^{[1]}, I_{1+2\delta}^{[1]})(1)||_1)$$

- Loss 2: This second constraint acts as a regularizer to maintain both spatial and temporal consistency across the entire sequence. It enforces similarity between the predictions from Stage II and the initial predictions from Stage I at corresponding timestamps.

$$LLC_2 = argmin_{M_\theta}\frac{1}{2}(||I_{2\delta}^{[1]} - I_{2\delta}^{[2]}||_1 + ||I_{1+\delta}^{[1]} - I_{1+\delta}^{[2]}||_1)$$

We then use the weighted sum of these losses as our final loss for our model.

$$\mathcal{L} = \lambda_{CC_1}\mathcal{L}_{CC_1} + \lambda_{CC_2}\mathcal{L}_{CC_2}$$

2.2 U-Net Backbone

The architecture is a U-Net (4), which allows for multi-scale feature extraction and precise spatial localization.

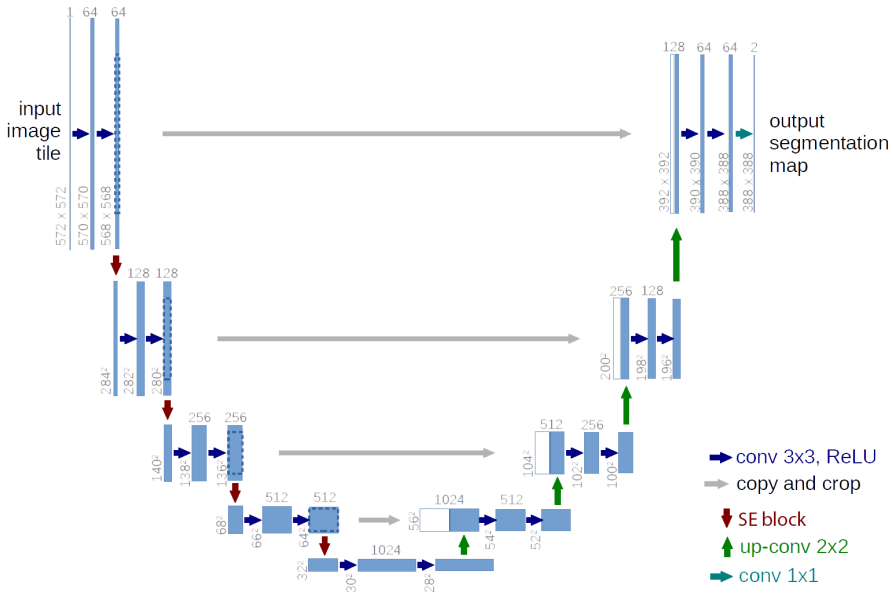


Figure 2.2: U-Net architecture. Source : (4)

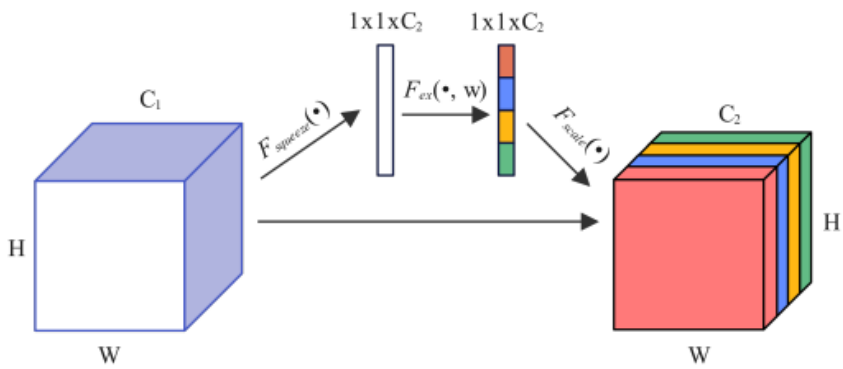


Figure 2.3: Squeeze-and-Excitation. Source : (2)

2.3 Squeeze-and-Excitation (SE) Blocks (2)

To enhance the representational power of the model, SE blocks were integrated in the U-Net structure. these blocks act as an attention mechanism, allowing the network to perform feature re-calibration. A Multi-Layer Perceptron (MLP) with a sigmoid activation assigns weights to each channel, emphasizing information features while suppressing less useful ones.

Chapter 3

Experimental Setup

3.1 Dataset Preparation

The dataset spans over ten years (2006-2017) and consists of daily-spaced temperature images.

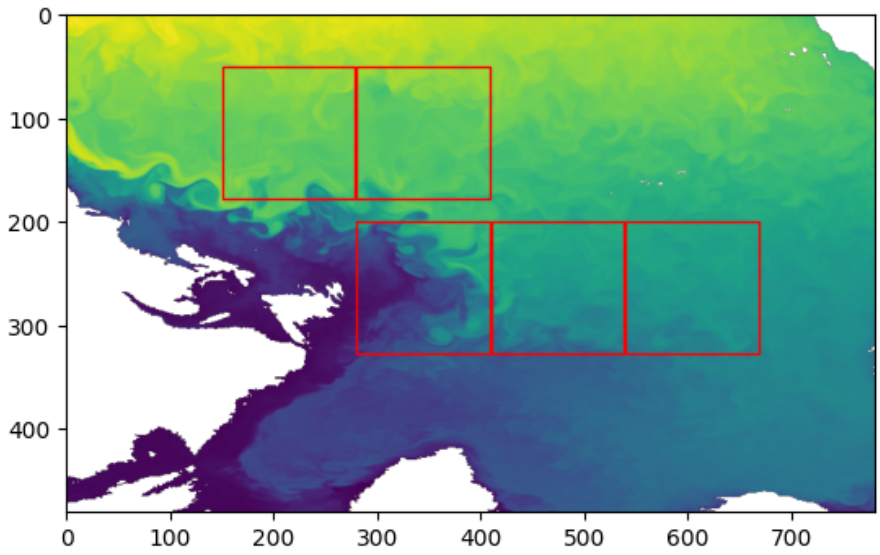


Figure 3.1: Sample of the dataset

- Spatial Patching: To maintain high training efficiency with available computational resources, the global imagery was subdivided into 128x128 pixel patches with 6 distinct patches extracted per image.

- Channel configuration: Each data point is represented as a single-channel grayscale image, where the pixel intensities correspond to normalized temperature values.

- Temporal Splitting: The data was partitioned into a training set, a 180-day validation set, and a 180-day test set to ensure an unbiased evaluation of the model's predictive capabilities.

3.2 Training Protocol

The model training was conducted in two distinct phases: an initial self-supervised pre-training followed by a comparative analysis stage.

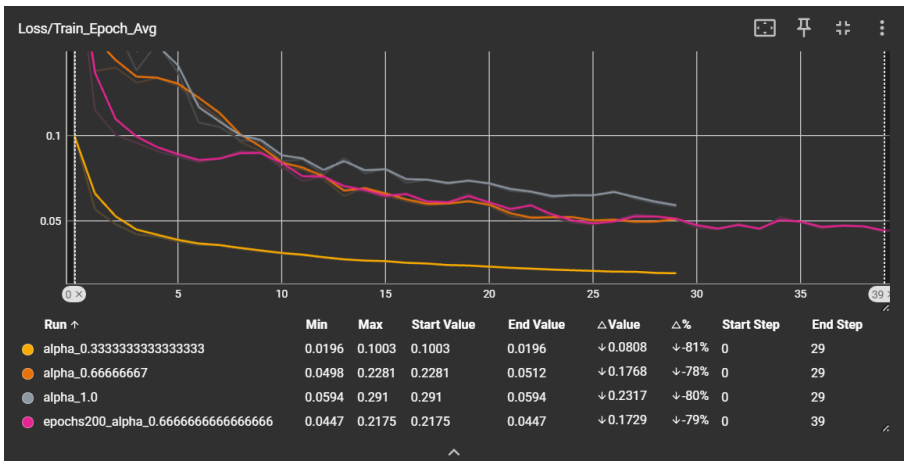
- Optimization: We employed the Adam optimizer to minimize the dual cycle consistency loss.
- Learning Rate: A fixed learning rate of 1×10^{-4} was maintained to ensure stable convergence.
- Training Duration: The network was trained for 30 epochs.

Chapter 4

Results and Discussion

4.1 Choice of Hyperparameter λ

A critical aspect of the training process is the balance between the two cycle consistency losses. As described in the optimization criteria ($\mathcal{L} = \lambda_1 \mathcal{L}_{CC_1} + \lambda_2 \mathcal{L}_{CC_2}$), we must assign relative weights to the reconstruction accuracy (λ_1) and the temporal stability regularizer (λ_2).



Run	Min	Max	Start Value	End Value	△Value	△%	Start Step	End Step
alpha_0.3333333333333333	0.0196	0.1003	0.1003	0.0196	↓0.0808	↓-81%	0	29
alpha_0.66666667	0.0498	0.2281	0.2281	0.0512	↓0.1768	↓-78%	0	29
alpha_1.0	0.0594	0.291	0.291	0.0594	↓0.2317	↓-80%	0	29
epochs200_alpha_0.6666666666666666	0.0447	0.2175	0.2175	0.0447	↓0.1729	↓-79%	0	39

Figure 4.1: Training curves for different lambda

To determine the optimal configuration, multiple experimental runs were conducted with varying values for these hyperparameters. We analyzed the average training loss per epoch across different ratios, including λ values of 0.33, 0.66, and 1.0.

The experimental results led to the following observations:

- Optimal Configuration: We selected $\lambda_1 = 1/3$ and $\lambda_2 = 2/3$ for our final model.

- Loss Performance: This specific configuration consistently yielded the lowest final training loss compared to other tested ratios.

4.2 Quantitative Analysis

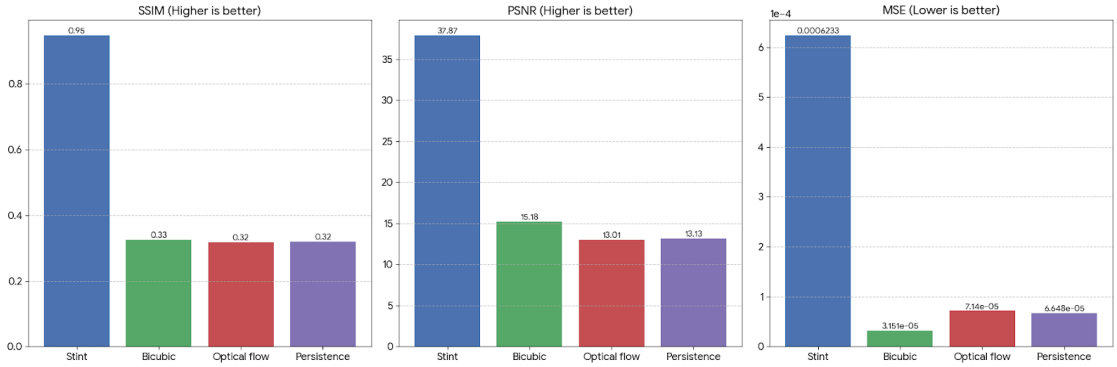


Figure 4.2: Comparison with baseline

The model was compared against three baselines using a synthetic temperature dataset. STint demonstrated a clear advantage in preserving the structural integrity of temperature fields.

- **Interpretation**: STint achieved a PSNR of 37.87 dB, significantly outperforming Bicubic (15.18 dB) and Optical Flow (13.01 dB). The high SSIM score of 0.95 confirms that the model accurately captures the spatial patterns and structural relationships within the temperature patches.
- **Failure of Traditional Methods**: The poor performance of Optical Flow (13.01 PSNR) in this domain underscores our problem statement: geospatial variables do not follow the rigid-body motion rules that standard flow algorithms expect.

4.3 Qualitative Demonstration

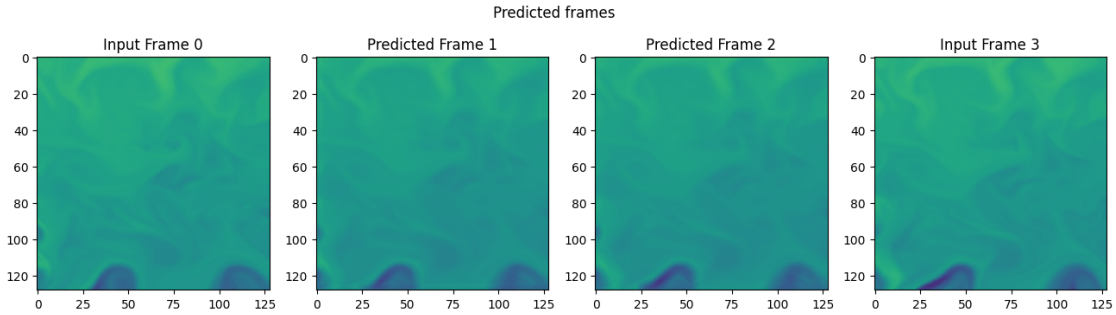


Figure 4.3: Sample of prediction. Frames 0 and 1 are the input, frames 1 and 2 are the predicted ones.

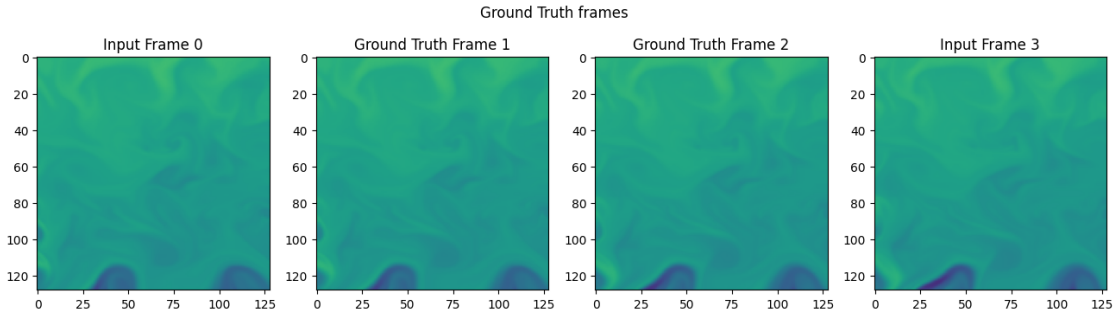


Figure 4.4: The corresponding ground truths.

Visual inspection of the predicted frames confirms the quantitative findings. While baseline methods often produced blurred results or simple average states, STint generated frames that were nearly indistinguishable from the ground truth.

4.4 Fine-tuning and Domain Transfer

To test the limitations of our flow-agnostic approach, the model was fine-tuned on the Sintel dataset, a high-frame-rate synthetic video dataset.

Figure 4.5: Sample of Sintel dataset

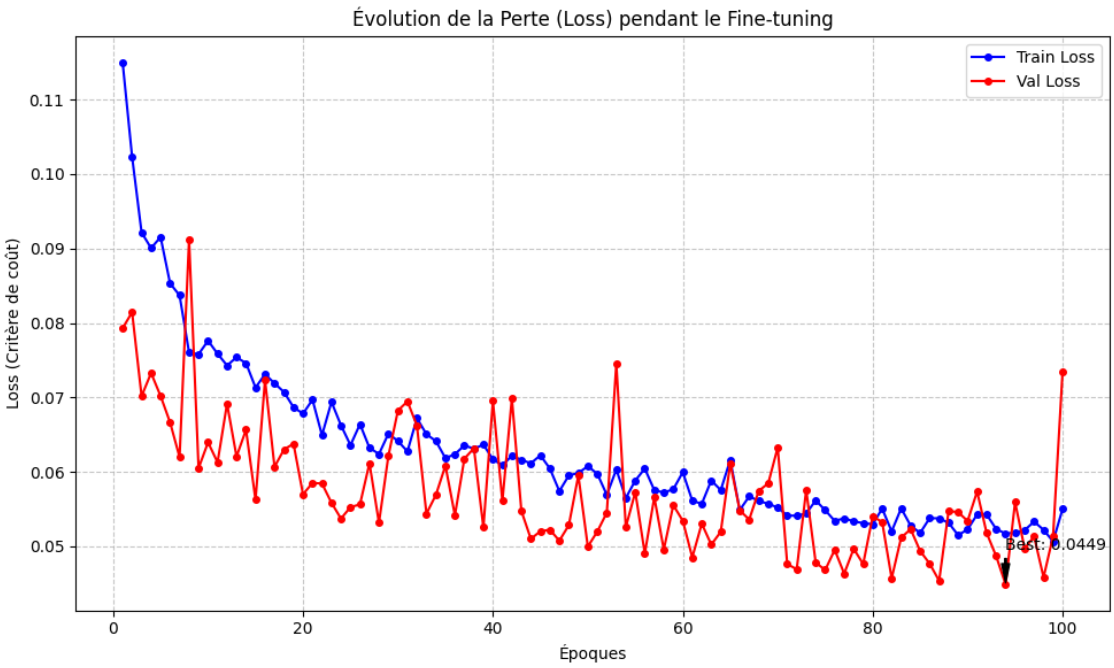


Figure 4.6: Learning curves during fine tuning

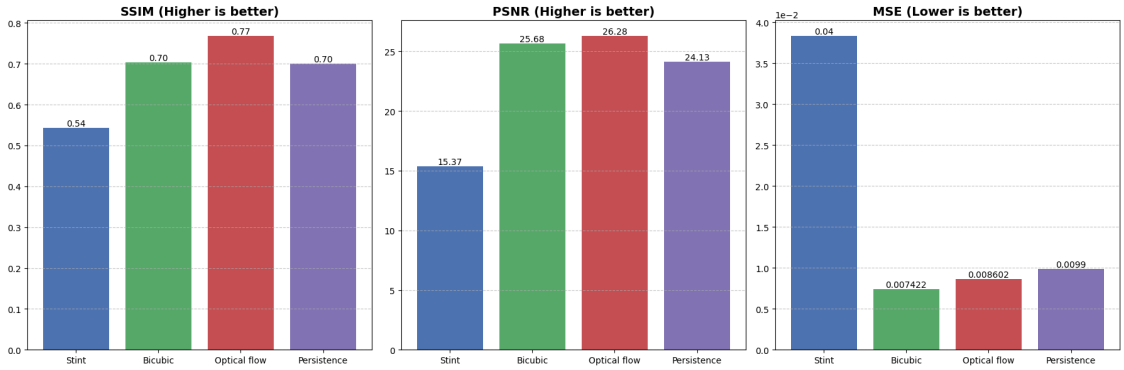


Figure 4.7: Comparison of fine tuning with baseline

- **Performance Drop**: Sintel, the PSNR dropped to 15.37 dB, and the SSIM to 0.54.

- **Discussion**: In this high-frame-rate environment, Optical Flow performed significantly better (PSNR 26.28). This highlights a key finding: STint is specialized for low-granularity, non-rigid data (like hours-long climate intervals) where flow assumptions break. In standard video where objects move predictably across pixels, motion-based methods remain superior.

4.5 Limitations

- **Training instability**: Training with cycle consistency is often prone to instability between batches.

- **Domain Specificity of Motion**: While the model excels at capturing non-rigid deformations like those in temperature or solar field data, it lacks the explicit motion information that remains highly effective for rigid-body interpolation in traditional computer vision.

- **Computational Efficiency**: The current multi-stage process is computationally demanding.

- **Performance on High-Frame-Rate Data**: Traditional optical flow-based methods like SuperSloMo (3) outperform the flow-agnostic approach.

Chapter 5

Conclusion

This technical report has detailed the implementation and evaluation of STint (Self-supervised Temporal Interpolation) as a robust solution for the unique challenges of geospatial data. By moving away from traditional optical flow assumptions, which often fail in the presence of low temporal resolution and non-rigid deformations, we have demonstrated a more physically consistent approach to temporal upsampling.
- **STint significantly outperforms traditional baselines** such as bicubic interpolation and optical flow in terms of PSNR and SSIM.
- **Flow-agnostic learning** is better suited for scientific variables like temperature, which undergo complex spatial deformations rather than rigid pixel-level movements.

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